MASTER RESEARCH (CPSE)

MP3: Industry 4.0: concept, survey and futur trends

« Flipped Classes »

Lionel Roucoules

25/09/2026

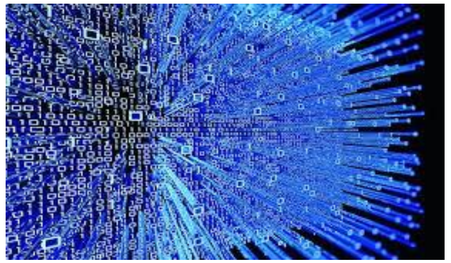


GOALS OF THE COURSE

- Discover the issues and technologies of industry 4.0
- Balance the use of these technologies in relation to uses







- Understand and apply research methodology

- Know how to use communication techniques, and know how to present concisely and in English
- Participate in a scientific exchange/debate





SESSIONS OF THE COURSE

Sessions 0(a) and 0(b)
- Introduction and first tutorial with coaching professors

Session #1 (5min, 5 slides max): introduction for an “informed” audience
- Introducing technology (1 slide)
- Introduce the usage and/or application (1 slide)
- Ask a research question with examples (1-2 slides)
- Synthesize as an image

Session #2 (7min, 7 slides max): current studies associated with the question
- What are the current studies on the subject?
- What are the technological and organizational solutions for their implementation?
- What are the current obstacles, and any ideas on how they would solve this?

Session #3 (7min, 7 slides max): experimental protocol
- Present an experiment (without doing it): a protocol and the data to answer the question and validate the path(s) proposed during session no. 2

Session n°4 (20min, 20 slides max): final presentation
- Sections presented during the first three sessions with the requested improvements

4 groups of 2 students
1 group of 3 students

Cross PGE and externals

Support your presentation with bibliographic references

ASSESSMENT METHODS

Interactions during sessions

- All sessions are rooms for exchanges between students.
- The final presentations should take into account the comments made during each of the presentations.
- The materials presented by the groups at the end of the 5th session are distributed to all students for review. This requires you to take notes during all sessions.

Assessment

- We note the different presentations and associated skills.
- The final test (knowledge of the content and the various questions relating to the presentations made) is noted.

PROPOSED THEMES

(USAGE AND TECHNOLOGY SITUATIONS)



DISTRIBUTION OF TOPICS

Topic 1: Artificial Intelligence for Object Recognition in CAD Models (JP. Pernot – SINCRONE Project)

Topic 2: Virtual and augmented reality for learning technical system (L. Roucoules – JENII project)

Topic 3: Cobotics to assist the assembly of systems on a production line (E. Yahia)

Topic 4: Digitalization (e- obeya, MES…) of workstations as support for assembly tasks (E. Yahia)

Topic 5: Graphic representation to support decision-making (L. Roucoules)

Topic 6: Digital twin for controlling the behavior of complex systems (F. Huet – Sphere project)

Topic 7 : Human and Artificial intelligence hybridation of manufacturing system supervision (K. Amzil – H-ARMON-IA project)

VIRTUAL AND AUGMENTED REALITY FOR VISUALIZATION AND INTERACTION WITH CPS

Advanced AR/VR: Human-System Interaction for:

- Better ergonomics
- Support for “business” tasks
- New individual and collaborative cognitive modes, in person or remotely









Collaboration between environments “Immersive” interactives

RA – RV Indiv . & coll.

MP Analysis

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n / testing
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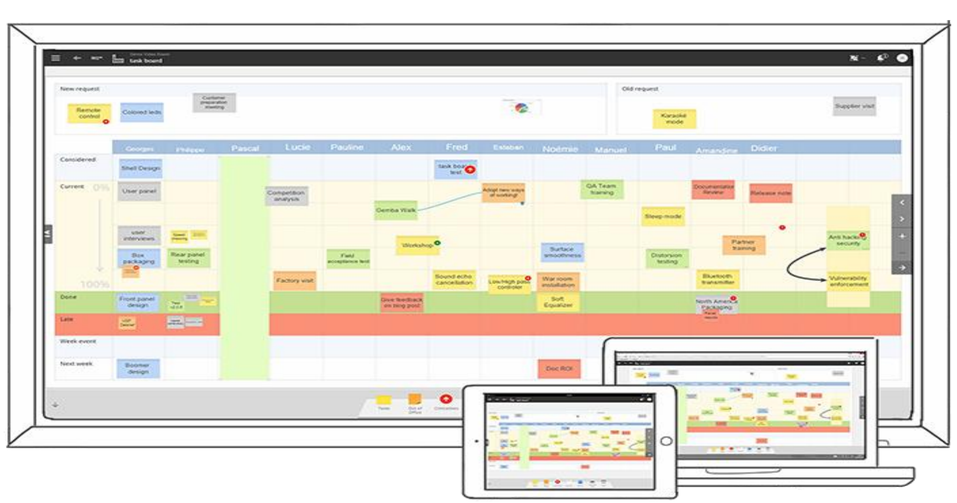
Digital twin



DEFINING KEY PERFORMANCE INDICATORS IN CPPS AND VISUALIZING THEM UNDER THE E- OBEYA

Digital visual management: real-time monitoring for:

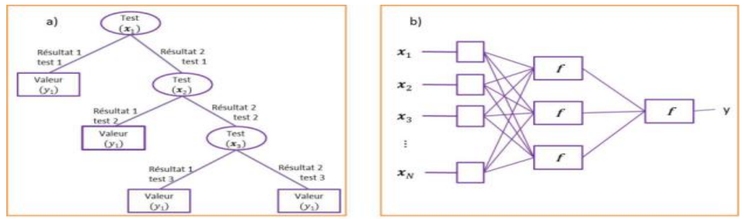
- Being at the most of real-time data (monitoring vs reporting)
- Share indicators across teams / Aggregate data to different indicators
- Helping decision making

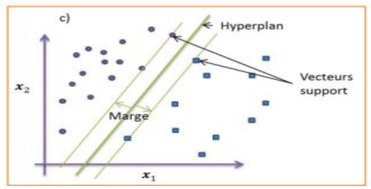




AI APPLIED TO CPS: APPLICATION TO THE LEARNING FACTORY

- CAD Object Recognition
- Preventive maintenance
- Interoperability of information systems





Upstream decision
Product Definition

RA – RV Indiv . & coll.

FAO

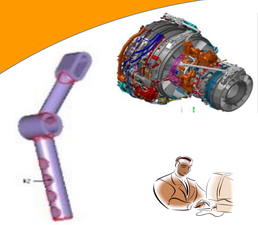
FEA

MP Analysis

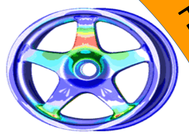
Conc . SP

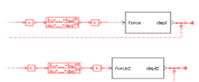
RA – RV Indiv . & coll.

Digital Engineering Chain
Collaborative
Modeling, simulation of behaviors and “ flat ” business representations)

















Collaboration between environments
“Immersive” interactives



Downstream decisions
Product validation / testing
System manufactured

Decision in use
System management/ supervision

Product Life / System





Modeling and simulation
Iot
RA – RV Indiv . & coll.

Digital twin

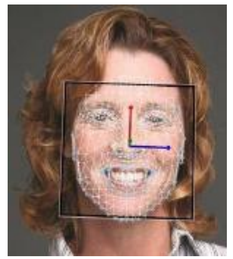


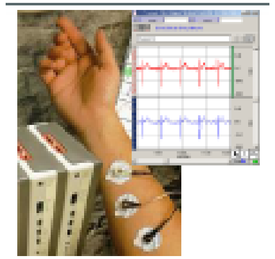


CONCEPTS AND ENABLING TECHNOLOGIES FOR THE DEVELOPMENT OF HUMAN CYBER-PHYSICAL SYSTEMS WORK

Digitalization for production systems







Upstream decision
Product Definition

RA – RV Indiv . & coll.

FAO

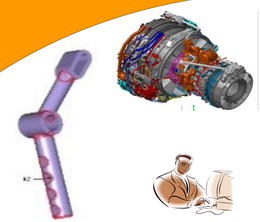
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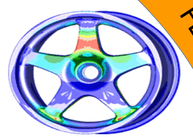
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COBOTIC FOR CYBER-PHYSICAL PRODUCTION SYSTEMS







Upstream decision
Product Definition

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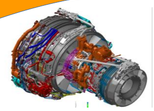
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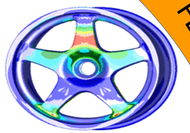
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